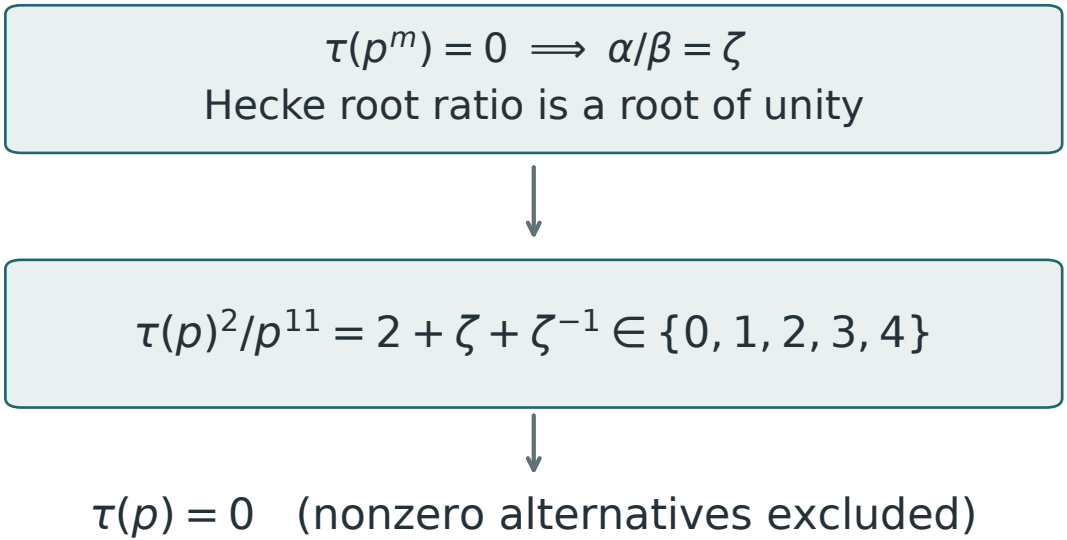
$\tau(p^m) = 0 \implies \alpha/\beta = \zeta$
Hecke root ratio is a root of unity
$\tau(p)^2/p^{11} = 2 + \zeta + \zeta^{-1} \in \{0, 1, 2, 3, 4\}$
$\tau(p) = 0$ (nonzero alternatives excluded)